

Computer lab 2 on Inference theory

1 Introduction

This computer lab consists of four parts:

1. Use of normal approximation/limit distribution results, for the construction of tests and confidence intervals.
2. Parametric and nonparametric tests
3. Regression analysis.
4. Research ethics

Try to answer all questions. Ask when you do not understand. You will analyse real data for the first two parts. For the third part you will read a guideline and answer questions related to that.

2 The use of normal approximation in testing and confidence interval construction

We will in this part construct a confidence interval for a functional $E(g(X))$ which is linear in F , when data come from the (unknown) distribution F . For this we will use the central limit theorem, the law of large numbers, the continuous mapping theorem and Slutsky's theorem. Convince yourself of how the statements of these theorems are used in the code below.

1. We generate 200 data points from some distribution, below we have used the $Exp(1)$ distribution, you may use this or choose any other distribution.

```
n<-200
x<-rexp(n)
```

We want to construct a confidence interval for the functional $\theta = E(e^{-X})$ (this is the moment generating function $\psi(t) = E(e^{tX})$ evaluated in $t = -1$). Note that this is a suggestion of a linear functional, where we have put $g(u) = e^{-u}$, you may use another functional if you wish. The code below uses the plug-in estimators of θ and of $Var(\theta)$. Convince yourself that you understand how.

```
teta.hat<-mean(exp(-x))
sigma.hat<-sd(exp(-x))
error<-qnorm(0.975)*sigma.hat/sqrt(n)
left<-teta.hat-error
right<-teta.hat+error
```

Then $(left, right)$ is a confidence interval for θ with the approximate confidence grade 0.95.

- (a) Calculate the theoretical value of θ for your chosen distribution and functional. (Note that this might not be possible for all choices, if that happens, choose some other distribution and/or functional).
- (b) Make a simulation study with $N = 1000$ data samples, analogously to what you have done previously, to check what the confidence intervals covering probability is.
- (c) For a fixed number $n = 200$, what does $count/N$ converge to? Use the law of large numbers to motivate your statement.
- (d) (Harder question!) If we let n instead vary: what does the theoretical covering probability of a test such as above converge to, when $n \rightarrow \infty$? Use the above mentioned theorems to motivate your statements.

3 Analysis of clinical research data

The data that you are supposed to analyse in this part of the lab come from the Clinical Research Center (CRC) at the University Hospital in Malmö (UMAS). The data are real data and are data that researchers at CRC have done (part of) their research on. Thus a main goal of this lab is to illustrate the types of analyses that one can do as a research scientist at a high quality research institute.

Note that there are no "right answers" to the questions below. You are supposed to analyse the material. You can of course do this in a more or less clever way. The lab instructions tell you what you should think of when you perform the analyses, but again there are no "right answers" to the research questions. And you can possibly find a new connection or result that previously was not known!

3.1 The data material

Data consists of $n = 4547$ individuals that have been followed until either (i) they die (in a heart disease) or until (ii) they leave the study, for some reason. For each individual there are measurements on a number of phenotypes

T2D
 FASTINGGLUCOSE
 CHOL
 TRIGL
 HDLCHOL
 LDL
 FASTINGINSULIN
 BMI
 WH
 SEX
 SMOKING
 PHYSACTIVITY

The phenotypes are measurements on

1. whether the individual is diabetic (T2D)
2. fasting blood sugar level (FASTINGGLUCOSE)
3. total cholesterol level (CHOL)
4. blood triglyceride level (TRIGL)
5. HDL cholesterol level, the "good" cholesterol (HDLCHOL)
6. HDL cholesterol level, the "bad" cholesterol (LDL)

7. fasting insulin level, high value indicates insulin insensitivity, which is "bad" (FASTINGINSULIN)
8. body mass index value, obesity measurement (BMI)
9. waist hip ratio, obesity measurement (WH)
10. the sex of the individual (SEX) value 1 is male
11. whether individual is a smoker (SMOKING) value 1 is smoker
12. the amount of physical activity (PHYSACTIVITY)

There are also, for each individual, measurements on 7 genotypes. These genes have actual names, known to the researchers at CRC, but for ethical purposes they are for us coded as gene 1, gene 2, ..., gene 7. In the data material we have they are labeled as

`g1, g2, g3, g4, g5, g6, g7`

The individuals are identified by two identifiers:

`PATIENT`
`FAMILY`

The identifier `PATIENT` is a unique identifier for each individual, while the identifier `FAMILY` can be shared by several individuals (that belong to the family identified by the variable `FAMILY`).

3.2 Load data into R

The data file "datafil.txt" has been sent to you by email. Before you can run the below command to load the data into R, you need to save the data file to a directory and change R's working directory to the directory in which you have saved the file "datafil.txt". After having done that you can run the below command to load the data.

To load the data into R's data working memory: Do

```
dat<-read.table("datafil.txt")
```

You can list all variables for the data by doing

```
names(dat)
```

4 Nonparametric tests and estimators

In this part you are supposed to get a picture of the distributions for the different phenotypes and do tests for if there is a difference between those distributions.

4.1 Nonparametric estimates

The empirical distribution function (ecdf) is an estimator of the true and unknown distribution function. To calculate the empirical distribution function, for instance BMI, do

```
plot(ecdf(dat$BMI))
```

If you want to look at the ecdf for BMI for the group that has the riskgenotype *g1* (corresponding to the value *g1* = 1), do

```
plot(ecdf(dat$BMI[dat$g1==1]))
```

If you want to see the ecdf for BMI for the two groups, that have and that do not have the risk genotype (corresponding to *g1* = 1 and *g1* = 0), in the same figure, do

```
plot(ecdf(dat$BMI[dat$g1==0]),xlim=c(15,60))
par(new=TRUE,col="red")
plot(ecdf(dat$BMI[dat$g1==1]),xlim=c(15,60))
```

Do this for some interesting groups. Some suggestions for possible group division and how you divide are: You can for instance divide into gender (variable SEX, 1 means man, 2 means woman), whether the individual is a smoker or not (variable SMOKING), those that have type II diabetes versus those that have not (T2D). Other interesting phenotypes apart from BMI that are continuous and that you can study the distribution for and distributional differences between groups for are CHOL, TRIGL, HDLCHOL, LDL, FASTINGINSULIN.

4.2 Nonparametric tests

The above gives you estimates of the distribution functions and the plotted figures can give you an indication of whether there are differences of the distributions between groups.

To make a formal test of differences between groups you can use a two-sample Kolmogorov-Smirnov test (R code function is *ks.test*). Do

```
plot(ecdf(dat$HDL[dat$SMOKING==0]),xlim=c(0,4))
par(new=TRUE,col='red')
plot(ecdf(dat$HDL[dat$SMOKING==1]),xlim=c(0,4))
ks.test(dat$HDL[dat$SMOKING==1],dat$HDL[dat$SMOKING==0])
```

Do the figures and the formal test agree?

You can use the ecdf to estimate quantiles, by taking the inverse of the ecdf at a fixed point. More convenient and what amounts to the same thing is to calculate the empirical quantiles. Convince yourself that you understand the equivalence of this and ask the teacher if it is not clear to you. As an example, if you want to estimate the lower 10% quantile for HDL in the group of smokers do

```
quantile(dat$HDL[dat$SMOKING==1], 0.1, na.rm=TRUE)
```

You can use a one-sample Kolmogorov-Smirnov test to check if data are Normal distributed.

```
ks.test(dat$HDL[dat$SMOKING==1], "pnorm")
plot(ecdf(dat$HDL[dat$SMOKING==1]),xlim=c(0,4))
t<-seq(0,4,by=0.01)
m<-mean(dat$HDL[dat$SMOKING==1],na.rm=TRUE)
v<-var(dat$HDL[dat$SMOKING==1],na.rm=TRUE)
par(new=TRUE,col="blue")
plot(t,pnorm(t,m,sqrt(v)),xlim=c(0,4))
```

A better way to graphically see if there is a difference to the Normal distribution is to use a qq-plot.

```
qqnorm(dat$HDL[dat$SMOKING==1])
```

Use the help function on "qqnorm", and convince yourself that you understand what a qq-plot does. Ask if you don't understand.

5 Nonparametric estimation of the probability density function

Load the package *KernSmooth* into R's program memory

```
library(KernSmooth)
```

Do help on the function *bkde*. Choose one of the variables that you could be interested in estimating the density for, and estimate the density for that variable, for instance for BMI the code is

```
fit<-bkde(dat$BMI[!is.na(dat$BMI)])
plot(fit,type="l")
```

Try to change the bandwidth to some different values.

```
fit<-bkde(dat$BMI[!is.na(dat$BMI)],band=1)
plot(fit,type="l")
fit<-bkde(dat$BMI[!is.na(dat$BMI)],band=0.5)
plot(fit,type="l")
fit<-bkde(dat$BMI[!is.na(dat$BMI)],band=0.3)
plot(fit,type="l")
fit<-bkde(dat$BMI[!is.na(dat$BMI)],band=0.1)
plot(fit,type="l")
fit<-bkde(dat$BMI[!is.na(dat$BMI)],band=0.05)
plot(fit,type="l")
```

What is your explanation to what you see?

Try to plot the kernel estimator of the density (for some choice of bandwidth) against a Gaussian density (for instance with values for the expectation and the variance estimated from the data), for a comparison. Explain what you see. Is this a formal test, and why not? Explain where in the lab you did the formal test.

6 Parametric tests

One can use for instance a t-test to test for whether there is a difference between the expectations for a covariate in different groups. Note that even if we see, as above, that the distributions are not Gaussian, since the estimates of the expectations are averages, and since an average of i.i.d. data is approximately Gaussian by the CLT, the t-test will still give (approximately) correct p-values.

Note however that the parametric estimates are not as informative for the (whole) distribution, since we only estimate the expectation in the distribution, and note also that parametric tests are not as sensitive for differences, since we only test for differences between the expectations in the distributions.

To do a t-test, do

```
t.test(dat$HDL[dat$SMOKING==1],dat$HDL[dat$SMOKING==0])
```

Try to do some more analyses. Do you get the same results as in the previous section, using non-parametric tests?

7 Linear regression

7.1 Univariate linear regression

We will use two variables as response variables, BMI and WH, and try to find what other variables might influence them, and try to estimate or describe that influence.

A univariate linear regression model can be fitted to the data as

```
fit<-lm(BMI~FASTINGGLUCOSE,data=dat)
summary(fit)
```

Give an interpretation of the results! Do help on *lm*. Try using other explanatory variables. To graphically see if the model fits data one can plot the residuals

```
plot(fit$fit,fit$res)
```

What should the residuals look like? Ask the teacher what you can do if they do not look like they should. To see if the residuals are Normal distributed, you can do a qq-plot

```
qqnorm(fit$res)
```

Formal test of if the model fits the data is given by the F-statistic in

```
summary(fit)
```

Ask if you do not understand.

7.2 Multivariate linear regression

A multivariate regression model can be fitted to the data (using the least squares method), for instance using the covariates FASTINGGLUCOSE, CHOL, TRIGL, HDLCHOL, LDL and FASTINGINSULIN as explanatory variables, by

```
fit<-lm(BMI~FASTINGGLUCOSE+CHOL+TRIGL+HDLCHOL+LDL+FASTINGINSULIN,data=dat)
summary(fit)
```

Interpret the results! If you compare this model with a model with only FASTINGGLUCOSE as explanatory variable, you will find that FASTINGGLUCOSE is not any more significant in the multivariate regression model. Why?

7.2.1 Model selection in multivariate regression

The question of which covariates you should end up using in a multivariate model, is known as a model selection problem. To do model selection in a regression model it is convenient to use the commands *add1* and *drop1*. One of the standard methods for that is to do so called stepwise backwards elimination of a "large" model, as long as you can. In the example above one would do

```
fit<-lm(BMI~FASTINGGLUCOSE+CHOL+TRIGL+HDLCHOL+LDL+FASTINGINSULIN,data=dat)
drop1(fit,test="Chisq")
fit<-lm(BMI~CHOL+TRIGL+HDLCHOL+LDL+FASTINGINSULIN,data=dat)
drop1(fit,test="Chisq")
etc
```

1. Use the above method to find a model for BMI. Start with a multivariate model that uses *all* variables as explanatory and eliminate them one at a time, until it is no longer possible.
2. Do the same thing (i.e. find a good model) for WH.
3. Find separate models for BMI (and for WH), for men and for women. What are your conclusions? To do an analysis on a subset of all data, for instance for only males, one can use the code

```
fit<-lm(BMI~FASTINGGLUCOSE+CHOL+TRIGL+HDLCHOL
+LDL+FASTINGINSULIN,data=dat,subset=SEX==1)
```

A way to incorporate a variable that you believe "should be included" but that was not significant, is to make a (non-linear) transformation of the variable. For instance one could dichotomise a continuous variable. Try to do this for FASTINGGLUCOSE, by dichotomising for instance at it's median

```
fast.m<-quantile(dat$FASTINGGLUCOSE,0.5,na.rm=TRUE)
dat$ny.fast<-as.real(dat$FASTINGGLUCOSE>fast.m)
fit<-lm(BMI~ny.fast+CHOL+TRIGL+HDLCHOL+LDL+FASTINGINSULIN,data=dat)
summary(fit)
```

What is your conclusion?

8 Research ethics

Read the full document "The European Code of Conduct for Research Integrity". Answer the following questions.

1. In Section 2.2 Training, Supervision and Mentoring, one point reads: "Researchers report their results and methods, including the use of external services or AI and automated tools, in a way that is compatible with the accepted norms of the discipline and facilitates verification or replication, where applicable." Do you believe that your written lab report conforms with this? Is yes, explain why. If not, explain what is lacking for it to conform to the point.
2. Which point in 2.5 Data Practices and Management do you personally consider to be the most important? Motivate.
3. In Section 2.7 Publication and Dissemination one point reads: "Authors acknowledge important work and contributions of those who do not meet the criteria for authorship, including collaborators, assistants, and funders who have enabled the research. ". Suppose that you would write a report on the analysis of data in this lab, but that you are a researcher in the medical lab facility, where the data has been gathered. Which categories of personnel would you need to include in the final report, to conform with this point?
4. Concerning 3.1 Research Misconduct and other Unacceptable Practices, there are three forms of violation listed: Fabrication, Falsification and Plagiarism. Suppose that you would when analysing the data in this lab first do a "pre-screening" on the data and e.g. do several tests to see which hypotheses may be interesting to investigate. Suppose then that you would do a statistical analysis of the selected hypotheses and (only) report the results of your analysis of these, without taking into consideration or reporting the "pre-screening". Which (if any) of the three forms have you violated?
5. Suppose that you are a young newly employed researcher working in a prestigious research institute, where the senior researcher, who has won numerous scientific awards, instructs you to exclude a part of the data that if they would be included lead to a covariate not being statistically significant in the analysis. Can you point to an item in the bulleted list in 3.1 Research Misconduct and other Unacceptable Practices, that explicitly addresses the violation?